Hyperspectral Unmixing Using Nonlocal Similarity-Regularized Low-Rank Tensor Factorization

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***Abstract*— Recently, methods based on nonnegative tensor factorization (NTF), which benefits from the tensor representation of hyperspectral imagery (HSI) without any information loss, have attracted increasing attention. However, most existing methods fail to explore the internal spatial structure of data, resulting in low unmixing performance. Moreover, when the algorithm is optimized, the solution is unstable. In this article, a regularizer based on nonlocal tensor similarity is proposed, which can not only fully preserve the global information of HSI but also mine the internal information of data in the spatial domain. HSI is regarded as a 3-D tensor and is directly subjected to endmember extraction and abundance estimation. To fully explore the structural characteristics of data, we simultaneously use the local smoothing and low tensor rank prior of the data to constrain the unmixing model. First, several 4-D tensor groups can be obtained after the nonlocal similarity structure of HSI is learned. Subsequently, a low tensor rank prior is applied to each 4-D tensor, which can fully simulate the nonlocal similarity in the image. In addition, total variation (TV) is also used to explore the local spatial relationship of data, which can generate a smooth abundance map through edge preservation. The optimization is solved by the ADMM algorithm. Experiments on synthetic and real data illustrate the superiority of the proposed method.**

***Index Terms*— Hyperspectral unmixing, low tensor rank, nonlocal similarity, nonnegative tensor factorization (NTF).**

I. Introduction

HYPERSPECTRAL imagery (HSI) is entering the field of vision of more and more researchers with superior ability to obtain spectral information [1] and has been successfully applied in various tasks [1]. However, due to the limitation of imaging conditions, pixels in HSI are generally composed of a mixture of several different materials. Therefore, unmixing is introduced into image processing tasks to improve the utility of HSI, which aims at extracting pure substances (i.e., endmembers) in the image and determining the proportional distribution (i.e., abundances) of each component.

In recent years, a variety of unmixing models have emerged. Among them, the performance of the linear spectral mixture model (LMM) is particularly outstanding, which assumes that the mixed pixels are formed by a linear combination of endmembers. Under this model, some methods consider that there exist pure pixels in the image, such as pixel purity index (PPI) [2], vertex component analysis (VCA) [3], and N-FINDR [4]. However, other methods hold different views. They believe that pixels are highly mixed in many scenes, such as mining areas. Nonnegative matrix factorization (NMF) is a popular method to solve this problem, which aims at finding a set of the basis for data and its expression coefficient under the basis [5]. Unfortunately, due to the nonconvexity of the objective function, NMF often obtains a local optimal solution. To accurately extract endmember and abundance information, researchers try to add various constraints to the NMF framework based on the known prior information to make the model more effective, such as sparsity and spatial constraints.

Since 3-D HSI is reshaped into a 2-D matrix in the NMF framework, it directly destroys the obvious positional relationship of pixels in the original space and also weakens the correspondence between spectral information and spatial information [6]. Therefore, researchers have spent a lot of energy trying to find some regularization, which can explore the spatial relationship between pixels to constrain the abundance matrix [7]–[12]. Liu *et al.* [7] show a spatial constraint, in which the abundance of each pixel is considered to be consistent with the average of the surrounding pixels abundances. A local neighborhood preserving regularization is introduced in [8], which believes that each pixel can be reconstructed by a linear combination of pixels in its neighborhood. Lu *et al.* [9] learn the global similarity structure of the data and established a manifold regularizer to force the corresponding abundance vectors to be similar. A hypergraph used to capture the empty spectrum structure of the data is also shown in [10]. Yang *et al.* [11] and Guan *et al.* [12], respectively, use manifold constraints to maintain the structure information of data in the abundance matrix. When using the NMF framework for unmixing, the methods introduced above can not only obtain a stable solution of NMF but also maintain the spatial information in the abundance matrix to a certain extent. Regrettably, even if many constraints are added, some

Manuscript received December 16, 2020; revised April 25, 2021; accepted May 23, 2021. Date of publication August 11, 2021; date of current version January 17, 2022. This work was supported in part by the National Key Research and Development Project under Grant 2020YFB2103902, in part by the National Science Fund for Distinguished Young Scholars under Grant 61825603, and in part by the Key Program of National Natural Science Foundation of China under Grant 61632018. *(Corresponding author: Yuan Yuan.)*

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Digital Object Identifier 10.1109/TGRS.2021.3095488

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spatial and spectral information inevitably are still lost since the added regularization is not enough to fully explore the spatial and spectral structures in the original HSI.

To overcome the limitations of NMF, some methods of directly analyzing HSI begin to be favored by researchers. An HSI cube can be regarded as a third-order tensor without any information loss. Therefore, treating unmixing as a nonnegative tensor factorization (NTF) problem is more prepared to describe the original mode of data, which is a kind of natural model. Compared with matrix, the tensor is more suitable for describing high-dimensional data, such as HSI and social networks [6]. Methods based on tensor analysis have been applied in many fields, such as image compression [13], [14], image segmentation [15], [16], denoising [17], [18], and discriminative analysis [19], [20]. In recent years, Dian *et al.* [21]–[25] propose a variety of data fusion methods based on tensors. The feature extraction method also uses the tensor model to obtain superior results [26], [27]. The common point of these methods is to use a low-rank tensor to simulate the high-order data and adds certain constraints to the low-rank expression model to achieve the purposes of accurately extracting data characteristics, reducing storage space, and removing noise.

According to the existing knowledge, Zhang *et al.* [28] apply tensor factorization (TF) to the unmixing for the first time, opening a new chapter in the extraction of endmembers and abundances based on tensor methods. Subsequently, CPD-based tensor analysis methods have been continuously tried to unmix [29], [30]. The CPD-based methods decompose the data into the sum of a finite number of tensors with rank 1, which is the rank of the original tensor. Unfortunately, this rank is usually much larger than the number of endmembers. As a result, it is difficult to associate each separation amount with endmembers and abundances after decomposition. Qian *et al.* [6] creatively introduce a matrix-vector NTF model, which decomposes a third-order tensor into multiple component tensors, and each component tensor can be expressed as the outer product of a matrix and a vector. This model greatly promotes the research progress of NTF on the unmixing task, and [31] and [32] are all the improvements made on it. Imbiriba *et al.* [33], [34] use low-rank tensor regularization to constrain abundance and account for spectral variability.

However, methods mentioned above regard HSI as a 3-D tensor, which not only ignores the internal structure of data and the continuous distribution characteristics of the ground objects but also fails to explore the interrelationship between data. According to [35], pixels in a local area are usually similar in the spatial domain. For an HSI cube, there are multiple local regions containing similar pixels that are similar to each other. A low-rank regularization is proposed in [36] to explore the intrinsic low-rank structure of data, in which the nuclear norm is used to recover the low-rank problem. Unfortunately, Huang *et al.* [36] need to adjust the weight parameter according to the size of the singular value, which has high computational complexity. Therefore, NTF-based hyperspectral unmixing methods still have plenty of room for improvement.

Based on the above prior knowledge, we propose a nonlocal low tensor rank (NLTR) regularization to constrain NTF unmixing framework in this article, so as to explore the spectral–spatial structure of raw data. First, an HSI represented by a 3-D tensor is divided into several tensor patches with overlapping according to specific window size and step sliding. The pixels in these small tensor patches mostly contain similar information. Then, patches are clustered together; each group can be constructed as a 4-D tensor containing two spatial modes: a spectral mode and a nonlocal mode. Since all pixels on the HSI share the same endmember matrix, the spatial structure relationship of HSI is transferred to the abundance tensor. Each 4-D tensor is composed of similar 3-D patches; then, a low-rank tensor regularization is applied to each clustered abundance tensor. Meanwhile, a global TV regularization is also added to the NTF framework to maintain the local smoothness of data space. Finally, we formulate the unmixing problem and solve it through the ADMM [37]. The complete flowchart is shown in Fig. 1. Specifically, our contributions in this article can be summarized as follows.

1) The nonlocal similarity of data is considered in this article. Through image segmentation and clustering, the 4-D tensor similarity groups are obtained. A regularization for low-rank constraint on 4-D tensor is then proposed, which can fully explore the nonlocal spatial information of data.
2) The TV regularizer describing the local smoothness of data combined with the nonlocal tensor low-rank regularization is used to constrain the NTF, which has a great advantage in maintaining the local spatial–spectral information of the data.
3) For the NTF-based unmixing model proposed, abundance nonnegative constraint (ANC) and abundance sum-to-one constraint (ASC) constraints are simultaneously performed, and alternating direction method of multipliers (ADMM) is used to solve. The experiments on synthetic and real data have proved the superiority and effectiveness of our proposed method.

The remaining organization of this article is as follows. The section II introduces some concepts and related definitions of tensor. In addition, related literature and corresponding expressions about unmixing used in this article are also given in detail. Our proposed method and its optimization are shown in detail in Section III. In Section IV, experimental results on synthetic and real data are presented. Section V gives the conclusions.

II. Related Works

In this section, we introduce in detail some basic concepts and operations, as well as linear mixed models for hyperspectral unmixing.

A. Notations

This section introduces some necessary notations that will be used later. As we all know, a tensor is an extension of a matrix, which is a multidimensional array, and the mode number of tensor represents its order [35]. Therefore, scalars

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are called zero-order tensors, vectors are first-order tensors, and matrices are second-order tensors. Here, we use an italic letter to represent a scalar, e.g., y and I. A bold lowercase letter is used to express vector, e.g., $\mathbf{y} \in \mathbb{R}^I$. A matrix is defined as a bold capital letter, e.g., $\mathbf{Y} \in \mathbb{R}^{I \times J}$. A tensor is given using a calligraphic letter, e.g., $\mathcal{Y} \in \mathbb{R}^{I_1, I_2, \ldots, I_N}$.

B. Concepts

This section describes some of the basic operations and definitions used in this article.

1) Tensor Mode: The different dimensions of a tensor are named modes. For example, an n-order tensor has n modes.

2) Frobenius Norm: The Frobenius norm of tensor $\mathcal{Y} \in \mathbb{R}^{I_1, I_2, \ldots, I_N}$ is defined as

$$||\mathcal{Y}||_F = \left(\sum_{i_1=1}^{I_1} \sum_{i_2=1}^{I_2} \cdots \sum_{i_n=1}^{I_N} |y_{i_1 i_2, \ldots, i_n}|^2 \right)^{1/2}. \tag{1}$$

3) l_1-Norm: The l_1-norm of tensor $\mathcal{Y}$ can be calculated by

$$\|\mathcal{Y}\|_1 = \sum_{i_1=1}^{I_1} \sum_{i_2=2}^{I_2} \cdots \sum_{i_n=1}^{I_N} \left| y_{i_1, i_2, \ldots, i_n} \right|. \tag{2}$$

4) Mode-n Unfolding: Mode unfolding refers to reshape a tensor into a corresponding matrix form. Given a third-order tensor $\mathcal{Y} \in \mathbb{R}^{I \times J \times K}$, its three matrix expansions are

$$\begin{aligned} \left(\mathbf{Y}_{(1)}\right)_{(j-1)K+k,i} &= y_{ijk} \\ \left(\mathbf{Y}_{(2)}\right)_{(k-1)I+i,j} &= y_{ijk} \\ \left(\mathbf{Y}_{(3)}\right)_{(i-1)J+j,k} &= y_{ijk}. \end{aligned} \tag{3}$$

5) Mode-n Product: The product of a tensor $\mathcal{A} \in \mathbb{R}^{I_1, I_2, \ldots, I_N}$ and a matrix $\mathbf{M} \in \mathbb{R}^{J \times I_N}$ can be expressed by the Tucker pattern-n product of the tensor $\mathcal{Y} = \mathcal{A} \times_n \mathbf{M}$. The details are given as follows:

$$(\mathcal{A} \times_n \mathbf{M})_{i_1, \ldots, i_{n-1} j i_{n+1}, \ldots, i_N} = \sum_{i_n=1}^{I_N} a_{i_1 i_2, \ldots, i_n} m_{j i_n}. \tag{4}$$

C. Linear Spectral Unmixing Model

In hyperspectral unmixing, the LMM means that all pixels in an HSI can be formed by linear combination of endmember spectra [38]. Specifically, given an observed HSI $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, the relationship between abundances and endmembers can be expressed as

$$\mathcal{Y} = \mathcal{A} \times_3 \mathbf{M} + \mathcal{N} \tag{5}$$

where $\mathbf{M} \in \mathbb{R}^{I_3 \times P}$ represents the endmember matrix, $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times P}$ denotes the abundance tensor corresponding to M, and $\mathcal{N} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ is the additional noise tensor. I_1, I_2, I_3, and P are height, width, the spectral bands, and the number of the endmembers in HSI.

According to [34], there are two constraints on the abundance tensor $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times P}$ in the function (4), i.e., abundance nonnegative constraint (ANC) and abundance sum-to-one constraint (ASC). ANC means that the contribution of each endmember to the pixel should not be less than zero, while ASC represents that the proportion of all endmembers in a pixel should be added to one. Therefore, the optimization problem of (4) can be written as

$$\begin{aligned} &J(\mathcal{A}, \mathbf{M}) = \frac{1}{2} \|\mathcal{Y} - \mathcal{A} \times_3 \mathbf{M}\|_F^2 \\ &\text{s.t. } \mathcal{A} \geq \mathbf{0}, \quad \mathcal{A} \times_1 \mathbf{1}_P = \mathbf{1}_{I_1 \times I_2} \end{aligned} \tag{6}$$

in which $\mathbf{1}_{I_1 \times I_2}$ represents a matrix whose elements are 1, $\mathcal{A}\times_1$ is the mode-1 unfolding of tensor $\mathcal{A}$, and $\mathbf{1}_P$ denotes a P-dimensional vector of all 1.

Unfortunately, although the above unmixing model regards HSI as a tensor, it ignores the spatial information. To fully consider the information contained in the HSI and assist in unmixing, a total variation (TV) regularization that connects the relationship between pixels around is introduced into the LMM to explore the local smooth structure of data [32], [39], [40]. Specifically, the TV regularization can be described as

$$\|\mathcal{A}\|_{\mathrm{TV}} = \|\mathcal{D}\mathcal{A}\|_1 = \|\mathcal{D}_h \mathcal{A}\|_1 + \|\mathcal{D}_v \mathcal{A}\|_1 \tag{7}$$

where $\mathcal{D}$ contains two operators, $\mathcal{D}_h$ and $\mathcal{D}_v$, which are used to represent the differentiation in the horizontal and vertical directions of the image spatial domain, respectively. $\|\cdot\|_1$ represents the l_1-norm. Assuming that (i_1, i_2, p) represents the coordinate of a pixel $\mathcal{A}(i_1, i_2, p)$ in the tensor, then the partial differential in horizontal and vertical directions can be expressed as

$$\begin{cases} \mathcal{D}_h \mathcal{A}(i_1, i_2, p) = \mathcal{A}(i_1, i_2 + 1, p) - \mathcal{A}(i_1, i_2, p) \\ \mathcal{D}_v \mathcal{A}(i_1, i_2, p) = \mathcal{A}(i_1 + 1, i_2, p) - \mathcal{A}(i_1, i_2, p). \end{cases} \tag{8}$$

Therefore, the unmixing model can be described as

$$\begin{aligned} &\min_{\mathcal{A}} \frac{1}{2} \|\mathcal{Y} - \mathcal{A} \times_3 \mathbf{M}\|_F^2 + \lambda_{\mathrm{TV}} \|\mathcal{A}\|_{\mathrm{TV}} \\ &\text{s.t. } \mathcal{A} \geq \mathbf{0}, \quad \mathcal{A} \times_1 \mathbf{1}_P = \mathbf{1}_{I_1 \times I_2} \end{aligned} \tag{9}$$

where λ_{TV} is the penalty parameter of TV regularization. However, since (7) only considers the partial smoothing among the data, it still has many problems. For example, not only the spatial and spectral information is not fully utilized but also the spatial structure of data has failed to be captured. In addition, the method mentioned above is also susceptible to noise. Therefore, we are committed to finding a new method for further exploration of spectral information and spatial information in this article.

III. Proposed Method

To explore the internal structure of data, we design a nonlocal low-rank regularization, combine it with the TV regularizer, and then add them to the NTF framework to form our algorithm, named NLTR. In this section, we introduce the proposed algorithm and optimization in detail.

A. Nonlocal Low-Rank Regularization

According to the given prior information, HSI is a 3-D cube data, which contains similar information in both the spatial and spectral domains. In the spectral domain, we assume in advance that all endmembers have no variability, and all pixels

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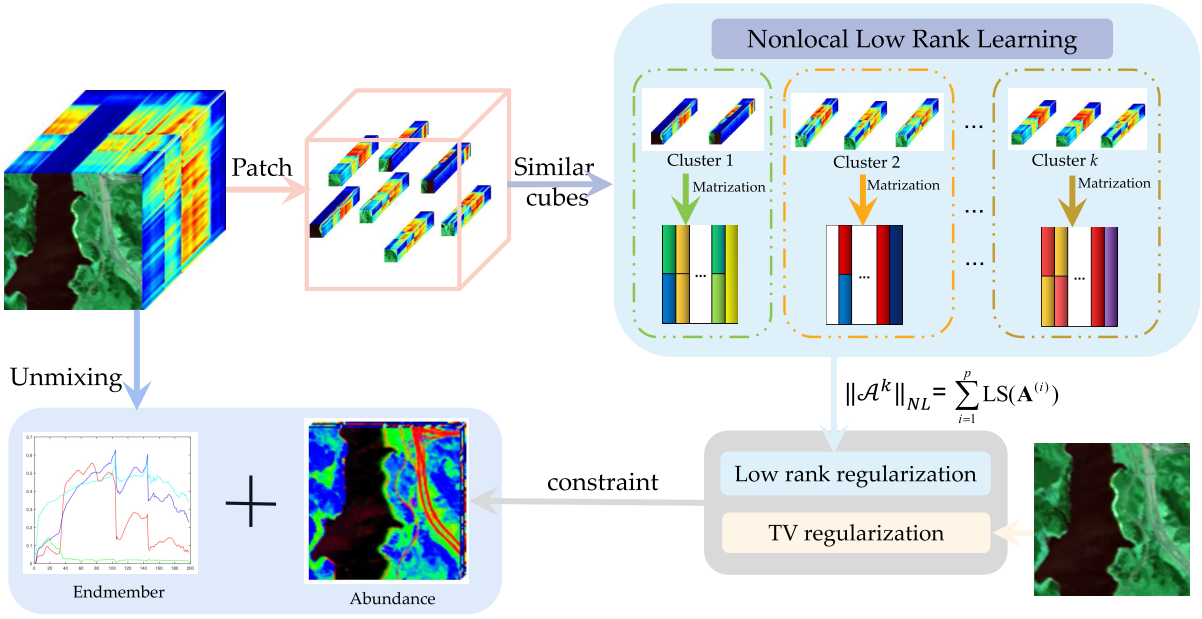


Fig. 1. Flow chart of NLTR.

share a spectral dictionary. At the same time, in the spatial domain, the pixels in the local area are similar, which contain the same or similar substances. For the entire HSI, there are many similar local areas, as shown in Fig. 1. Therefore, HSI not only has local similarity but also has nonlocal similarity. For this reason, many researchers have begun to explore similar relationships for unmixing tasks. In [35], the similarity relationship between the abundance maps is explored by a low-rank constraint. Inspired by this, we propose a nonlocal tensor regularization based on the NTF framework via using the nonlocal similarity property. The advantage is that a 3-D tensor can store spatial and spectral information as a whole, and the internal structure of data can be expressed by nonlocal low-rank regularization.

Based on the NTF model (9), the purpose of unmixing is to estimate the endmember matrix $\mathbf{M}$ and the abundance tensor $\mathcal{A}$ simultaneously. Combined with the above description, we rewrite the objective function as

$$\begin{aligned} &\min_{\mathcal{A}} \frac{1}{2}\|\mathcal{Y} - \mathcal{A} \times_3 \mathbf{M}\|_F^2 + \lambda_{\mathrm{TV}}\|\mathcal{A}\|_{\mathrm{TV}} + \lambda_{\mathrm{NL}}\|\mathcal{A}\|_{\mathrm{NL}} \\ &\text{s.t. } \mathcal{A} \geq \mathbf{0}, \quad \mathcal{A}\times_1 \mathbf{1}_P = \mathbf{1}_{I_1 \times I_2} \end{aligned} \tag{10}$$

in which $\|\mathcal{A}\|_{\mathrm{NL}}$ is a nonlocal low-rank regularization proposed in this article, and λ_{NL} is the penalty parameter.

Specifically, to explore the spatial–spectral structure of data, a low-rank regularization is designed to constrain the abundance tensor $\mathcal{A}$. First, in the spatial domain, a window with the size of $q \times q$ is used to split HSI with overlap, and many full-band patches can be obtained. Subsequently, the K-means++ algorithm [41] is used to cluster the divided patches. Thus, we can get K groups $Y^{(k)} = \{\mathcal{Y}^{(k,j)}\}_{j=1}^{N_k}$ after clustering, where $k = 1, 2, \ldots, K$, and N_k represents the number of patches in the kth clustering group.

Based on the prior knowledge, all pixels share the same endmember matrix, so image $\mathcal{Y}$ and abundance $\mathcal{A}$ have the same spatial similarity attributes. Here, we transfer the nonlocal information after the above segmentation to the abundance tensor. It can be obtained: $A^{(k)} = \{\mathcal{A}^{(k,j)} \in {}^{q\times q\times I_3}\}_{j=1}^{N_k}$. Finally, a 4-D tensor representing the nonlocal information of HSI can be built: $\mathcal{A}^{(k)} = \{\mathcal{A}^{(k)} \in {}^{q\times q\times I_3\times N_k}\}$, which has four modes, two spatial modes, one spectral mode, and one nonlocal mode. Within each 4-D tensor, there are many similar patches, which have extremely high correlations; hence, each tensor must be low rank. Embedding this prior property into (10), the objective function can be expressed as

$$\begin{aligned} &\min_{\mathcal{A}} \frac{1}{2}\|\mathcal{Y} - \mathcal{A}\times_3 \mathbf{M}\|_F^2 + \lambda_{\mathrm{TV}}\|\mathcal{A}\|_{\mathrm{TV}} + \lambda_{\mathrm{NL}} \sum_{k=1}^{K} \|\mathcal{A}^k\|_{\mathrm{NL}} \\ &\text{s.t. } \mathcal{A} \geq \mathbf{0}, \quad \mathcal{A}\times_1 \mathbf{1}_P = \mathbf{1}_{I_1 \times I_2}. \end{aligned} \tag{11}$$

For a matrix $\mathbf{A}$, the nuclear norm used to replace the rank constraint is usually represented by the sum of singular values, i.e., $\|\mathbf{A}\|_* = \sum_i \sigma_i(\mathbf{A})$, where $\sigma_j(\mathbf{A})$ is the ith singular value of $\mathbf{A}$. Since the nuclear norm of matrix often needs to be adjusted for the singular value parameters, a more effective method for solving the matrix low-rank problem is proposed in [42], which is given as

$$\mathrm{LS}(\mathbf{A}) = \sum_i \log(\sigma_i(\mathbf{A}) + \varepsilon) \tag{12}$$

in which ε denotes a very small positive. In this article, we define the nonlocal low-rank regularization $\lambda_{\mathrm{NL}}\|\mathcal{A}\|_{\mathrm{NL}}$ of tensor $\mathcal{A}$ by the following formula:

$$\|\mathcal{A}^k\|_{\mathrm{NL}} = \sum_{i=1}^{p} \mathrm{LS}(\mathbf{A}^{(i)}) \tag{13}$$

where $\mathbf{A}^{(i)} \in R^{(q\times q)\times N_k}$ is the ith slice of the tensor $\mathcal{A}^k$ along the spectral direction. Finally, the objective function that we

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proposed is expressed as

$$\begin{aligned}\min_{\mathcal{A}} \ & \frac{1}{2}\|\mathcal{Y}-\mathcal{A}\times_3\mathbf{M}\|_F^2+\lambda_{\mathrm{TV}}\|\mathcal{A}\|_{\mathrm{TV}} \\ & +\lambda_{\mathrm{NL}}\sum_{k=1}^{K}\sum_{i=1}^{I_3}\mathrm{LS}\left(\mathbf{A}_{(k)}^{i}\right) \\ \text{s.t. } & \mathcal{A}\geq\mathbf{0},\quad \mathcal{A}\times_1\mathbf{1}_P=\mathbf{1}_{I_1\times I_2}. \end{aligned} \tag{14}$$

B. Optimization

Formula (14) is an optimization problem with two constraints, which is more difficult to solve directly. Here, we use ADMM to solve, which is transformed into many subproblems that can be solved. Therefore, (14) can be transformed to

$$\begin{aligned}\min_{\mathcal{A},\mathbf{M}} \ & \frac{1}{2}\|\mathcal{G}_1-\mathcal{Y}\|_F^2+\lambda_{\mathrm{TV}}\|\mathcal{G}_3\|_1+\lambda_{\mathrm{NL}}\|\mathcal{G}_4\|_{\mathrm{NL}} \\ & +l_{R^+}(\mathcal{G}_5)+l_{\{1\}}(\mathcal{G}_6) \\ \text{s.t. } & \mathcal{G}_1=\mathcal{A}_{\times_3}\mathbf{M} \\ & \mathcal{G}_2=\mathcal{A} \\ & \mathcal{G}_3=\mathcal{D}\mathcal{G}_2 \\ & \mathcal{G}_4=\mathcal{A} \\ & \mathcal{G}_5=\mathcal{A} \\ & \mathcal{G}_6=\mathcal{A} \end{aligned} \tag{15}$$

in which $\mathcal{G}_1$, $\mathcal{G}_2$, $\mathcal{G}_3$, $\mathcal{G}_4$, $\mathcal{G}_5$, and $\mathcal{G}_6$ are intermediate variables, aiming at transforming the complex model into simple subproblems. $\mathcal{G}_5$ is used to meet ANC, which can ensure that $\mathcal{A}$ is not less than 0 during the iteration, that is,

$$l_{R^+}(\mathcal{A})=\begin{cases}0, & \text{if } \mathcal{A}\geq 0\\ +\infty, & \text{otherwise.}\end{cases} \tag{16}$$

The calculation method of $\mathcal{G}_6$, which is used to constrain the ASC, is defined as

$$l_{\{\mathbf{1}\}}(\mathcal{A})=\begin{cases}0, & \text{if } \mathrm{sum}(\mathcal{A}_{\times_3})=\mathbf{1}\\ +\infty, & \text{otherwise}\end{cases} \tag{17}$$

where $\mathbf{1}_{I_1\times I_2}$ represents a matrix whose elements are 1.

Therefore, the Lagrangian function of problem (15) can be written as

$$\begin{aligned}&\mathcal{L}(\mathcal{A},\mathbf{M},\mathcal{G}_1,\mathcal{G}_2,\mathcal{G}_3,\mathcal{G}_4,\mathcal{G}_5,\mathcal{G}_6,\mathcal{H}_1,\mathcal{H}_2,\mathcal{H}_3,\mathcal{H}_4,\mathcal{H}_5,\mathcal{H}_6) \\ &\quad=\min \frac{1}{2}\|\mathcal{G}_1-\mathcal{Y}\|_F^2+\lambda_{\mathrm{TV}}\|\mathcal{G}_3\|_1+\lambda_{\mathrm{NL}}\|\mathcal{G}_4\|_{\mathrm{NL}} \\ &\qquad+l_{R^+}(\mathcal{G}_5)+l_{\{\mathbf{1}\}}(\mathcal{G}_6)+\frac{\mu}{2}\|\mathcal{G}_1-\mathcal{A}\times_3\mathbf{M}+\mathcal{H}_1\|_F^2 \\ &\qquad+\frac{\mu}{2}\|\mathcal{G}_2-\mathcal{A}+\mathcal{H}_2\|_F^2+\frac{\mu}{2}\|\mathcal{G}_3-\mathcal{D}\mathcal{G}_2+\mathcal{H}_3\|_F^2 \\ &\qquad+\frac{\mu}{2}\|\mathcal{G}_4-\mathcal{A}+\mathcal{H}_4\|_F^2+\frac{\mu}{2}\|\mathcal{G}_5-\mathcal{A}+\mathcal{H}_5\|_F^2 \\ &\qquad+\frac{\mu}{2}\|\mathcal{G}_6-\mathcal{A}+\mathcal{H}_6\|_F^2 \end{aligned} \tag{18}$$

where $\mathcal{H}_1$, $\mathcal{H}_2$, $\mathcal{H}_3$, $\mathcal{H}_4$, $\mathcal{H}_5$, $\mathcal{H}_6$, and μ are Lagrangian multipliers and Lagrangian penalty parameter, respectively. For (18), according to the ADMM method, all variables $\mathcal{A}$, $\mathbf{M}$, $\mathcal{G}_1$, $\mathcal{G}_2$, $\mathcal{G}_3$, $\mathcal{G}_4$, $\mathcal{G}_5$, and $\mathcal{G}_6$ are solved in iterations. In the following, we will detail the update process of these variables.

1) All variables except $\mathbf{M}$ are related to $\mathcal{A}$; therefore, $\mathbf{M}$ is updated first. At this time, only $\mathbf{M}$ in (15) is an unknown variable, and it is a matrix, so there is the following update rule:

$$\mathbf{M}\leftarrow\mathbf{M}.*\left(\mathcal{Y}\times_3\mathcal{A}\times_3{}^{\mathrm{T}}\right)./\left(\mathbf{M}\mathcal{A}\times_3\mathcal{A}\times_3{}^{\mathrm{T}}\right). \tag{19}$$

2) For $\mathcal{A}$, problem (18) can be expressed as

$$\begin{aligned}\mathcal{L}(\mathcal{A})=\min_{\mathcal{A}} \ & \frac{\mu}{2}\|\mathcal{G}_1-\mathcal{A}\times_3\mathbf{M}+\mathcal{H}_1\|_F^2 \\ & +\frac{\mu}{2}\|\mathcal{G}_2-\mathcal{A}+\mathcal{H}_2\|_F^2 \\ & +\frac{\mu}{2}\|\mathcal{G}_4-\mathcal{A}+\mathcal{H}_4\|_F^2 \\ & +\frac{\mu}{2}\|\mathcal{G}_5-\mathcal{A}+\mathcal{H}_5\|_F^2 \\ & +\frac{\mu}{2}\|\mathcal{G}_6-\mathcal{A}+\mathcal{H}_6\|_F^2. \end{aligned} \tag{20}$$

It is obvious that (20) is a convex function, and it is easy to solve. Therefore, by taking the derivative of (20) and making it equal to zero, the update rule of tensor $\mathcal{A}$ can be obtained

$$\begin{aligned}\mathcal{A}\leftarrow & ((\mathcal{G}_1+\mathcal{H}_1)\times_3\mathbf{M}^T+\mathcal{G}_2+\mathcal{H}_2+\mathcal{G}_4+\mathcal{H}_4 \\ & +\mathcal{G}_5+\mathcal{H}_5+\mathcal{G}_6+\mathcal{H}_6)\left(\mathbf{M}^T\mathbf{M}+4\mathbf{I}\right)^{-1} \end{aligned} \tag{21}$$

in which $\mathbf{I}$ denotes the identity matrix.

3) Then, update six intermediate variables $\mathcal{G}_1$, $\mathcal{G}_2$, $\mathcal{G}_3$, $\mathcal{G}_4$, $\mathcal{G}_5$, and $\mathcal{G}_6$, and the objective function (18) becomes as follows:

$$\mathcal{G}_1=\arg\min_{\mathcal{G}_1}\ \frac{1}{2}\|\mathcal{G}_1-\mathcal{Y}\|_F^2+\frac{\mu}{2}\|\mathcal{G}_1-\mathcal{A}\times_3\mathbf{M}+\mathcal{H}_1\|_F^2 \tag{22}$$

$$\mathcal{G}_2=\arg\min_{\mathcal{G}_2}\ \frac{\mu}{2}\|\mathcal{G}_2-A+H_2\|_F^2+\frac{\mu}{2}\|\mathcal{G}_3-\mathrm{D}\mathcal{G}_2+H_3\|_F^2 \tag{23}$$

$$\mathcal{G}_3=\arg\min_{\mathcal{G}_3}\ \lambda_{\mathrm{TV}}\|\mathcal{G}_3\|_1+\frac{\mu}{2}\|\mathcal{G}_3-\mathcal{D}G_2+\mathcal{H}_3\|_F^2 \tag{24}$$

$$\mathcal{G}_4=\arg\min_{\mathcal{G}_4}\ \lambda_{\mathrm{NL}}\|\mathcal{G}_4\|_{\mathrm{NL}}+\frac{\mu}{2}\|\mathcal{G}_4-\mathcal{A}+\mathcal{H}_4\|_F^2 \tag{25}$$

$$\mathcal{G}_5=\arg\min_{\mathcal{G}_5}\ l_{R^+}(\mathcal{G}_5)+\frac{\mu}{2}\|\mathcal{G}_5-\mathcal{A}+\mathcal{H}_5\|_F^2 \tag{26}$$

$$\mathcal{G}_6=\arg\min_{\mathcal{G}_6}\ l_{\{\mathbf{1}\}}(\mathcal{G}_6)+\frac{\mu}{2}\|\mathcal{G}_6-\mathcal{A}+\mathcal{H}_6\|_F^2. \tag{27}$$

In detail, (22) is also a convex function, and its solution is similar to (21). Then, we can easily get the following solution:

$$\mathcal{G}_1\leftarrow\frac{1}{1+\mu}[\mathcal{Y}+\mu(\mathcal{A}\times_3\mathbf{M}-\mathcal{H}_1)]. \tag{28}$$

Since (23) contains a spatial convolution operator $\mathcal{D}$, we first find the partial derivative of (23) to $\mathcal{G}_2$. Then, the derivative is set to zero; we can get

$$\left(\mathcal{D}^T\mathcal{D}+\mathcal{I}\right)\mathcal{G}_2=\left(\mathcal{A}-\mathcal{H}_2+\mathcal{D}^T(\mathcal{G}_3+\mathcal{H}_3)\right) \tag{29}$$

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in which $\mathcal{I}$ is an identity tensor. According to the fast Fourier transformation (FFT) described in [43], the solution of (23) can be expressed as follows:

$$\mathcal{G}_2 \leftarrow \text{ifft}\left(\frac{\mathcal{A}-\mathcal{H}_2+\mathcal{D}^T(\mathcal{G}_3+\mathcal{H}_3)}{1+\text{fft}(\mathcal{D}_h)^T\text{fft}(\mathcal{D}_h)+\text{fft}(\mathcal{D}_v)^T\text{fft}(\mathcal{D}_v)}\right) \quad (30)$$

where $\mathcal{D}_h$, $\mathcal{D}_v$, fft, and ifft represent horizontal differentiation, vertical differentiation, Fourier transform, and inverse Fourier transform, respectively.

The solution of TV constraint $\mathcal{G}_3$ in (24) can be solved via the soft threshold [44]

$$\mathcal{G}_3 \leftarrow \text{soft}\left(\mathcal{H}_3-\mathcal{D}\mathcal{G}_2, \frac{\lambda_{\text{TV}}}{\mu}\right) \quad (31)$$

where soft is a threshold operator and defined as $\text{soft}(a,b) \equiv \text{sign}(a)\max\{|a|-b, 0\}$.

Equation (25) is the objective function with only the nonlocal similarity constraint proposed in this article, which can be rewritten as

$$\mathcal{G}_4 = \underset{\mathcal{G}_4}{\text{argmin}}\ \mu\|\mathcal{G}_4-\mathcal{A}+\mathcal{H}_4\|_F^2+\lambda_{\text{NL}}\sum_{k=1}^{K}\left\|\mathcal{G}_4^k\right\|_{\text{NL}}. \quad (32)$$

Then, (32) is divided into subproblems as follows:

$$\underset{\mathcal{G}_4^k}{\text{argmin}}\sum_{k=1}^{K}\mu\left\|\mathcal{G}_4^k-\mathcal{A}^k+\mathcal{H}_4^k\right\|_F^2+\lambda_{\text{NL}}\sum_{k=1}^{K}\left\|\mathcal{G}_4^k\right\|_{\text{NL}} \quad (33)$$

in which $\mathcal{G}_4^k$, $\mathcal{A}^k$, and $\mathcal{H}_4^k$ are the kth patches of tensors $\mathcal{G}_4$, $\mathcal{A}$, and $\mathcal{H}_4$, respectively. By introducing (13) into the above formula, it can get the following function:

$$\underset{\overline{\mathcal{G}}_4^k}{\text{argmin}}\frac{\mu}{q^2}\left\|\overline{\mathcal{G}}_4^k-\overline{\mathcal{V}}^k\right\|_F^2+\lambda\sum_{i=1}^{q^2}\text{LS}\left(\overline{\mathcal{G}}_4^k(:,\ :,\ i)\right) \quad (34)$$

where $\overline{\mathcal{G}}_4^k$ and $\overline{\mathcal{V}}^k$ are the FFT results of $\mathcal{G}_4^k$ and $\mathcal{A}^k-\mathcal{H}_4^k$ in the third mode, respectively. Specifically, they are defined as

$$\overline{\mathcal{G}}_4^k = \text{fft}\left(\mathcal{G}_4^k, [\ ], 3\right), \quad (35)$$

$$\overline{\mathcal{V}}^k = \text{fft}\left(\mathcal{A}^k-\mathcal{H}_4^k, [\ ], 3\right). \quad (36)$$

Subsequently, each slice matrix $\overline{\mathcal{V}}^k(:,\ :,\ i)$ is subjected to SVD, and we can get

$$\overline{\mathcal{V}}^k(:,\ :,\ i) = \mathbf{U}_1\Sigma_1\mathbf{V}_1. \quad (37)$$

Here, $\mathbf{U}_1$, Σ_1, and $\mathbf{V}_1$ obtained are used to solve $\overline{\mathcal{G}}_4^k(:,\ :,\ i)$ [25]. The solution of $\overline{\mathcal{G}}_4^k(:,\ :,\ i)$ can be constructed by

$$\overline{\mathcal{G}}_4^k(:,\ :,\ i) = \mathbf{U}_1\tilde{\Sigma}_1\mathbf{V}_1 \quad (38)$$

where $\tilde{\Sigma}_1$ is obtained by the threshold transformation operation of Σ_1, which can be expressed as follows:

$$\tilde{\Sigma}_1(i,i) = \text{E}_{\lambda_{\text{NL}},\varepsilon}[\Sigma_1(i,i)],$$
$$\text{E}_{\alpha,\varepsilon}(x) = \begin{cases} 0, & c_2 \le 0 \\ \dfrac{c_1+\sqrt{c_2}}{2}, & c_2 > 0 \end{cases} \quad (39)$$

in which c_1 and c_2 are given by

$$c_1 = |x|-\varepsilon$$
$$c_2 = c_1^2-4(\alpha-\varepsilon|x|). \quad (40)$$

Finally, the solution of $\mathcal{G}_4^k(:,\ :,\ i)$ can be obtained by the inverse FFT of $\overline{\mathcal{G}}_4^k(:,\ :,\ i)$, and the expression is given as follows:

$$\mathcal{G}_4^k \leftarrow \text{ifft}\left(\overline{\mathcal{G}}_4^k,\ [\ ],\ 3\right). \quad (41)$$

The complete process of updating $\mathcal{G}_4$ is summarized in Algorithm 1.

Algorithm 1 Overall Update Process of $\mathcal{G}_4^k$

Input: $\mathcal{G}_4^k$, $\mathcal{A}^k$, $\mathcal{H}_4^k$, λ_{NL}
Output: $\mathcal{G}_4$
compute $\mathcal{V}^k = \mathcal{A}^k-\mathcal{H}_4^k$; $\overline{\mathcal{V}}^k = fft\left(\mathcal{A}^k-\mathcal{H}_4^k, [\ ], 3\right)$
while $i \le q$ **do**
  $[\mathbf{U}_1, \Sigma_1, \mathbf{V}_1] = svd(\overline{\mathcal{V}}^k(:,\ :,\ i))$;
  $\tilde{\Sigma}_1(i,i) = \text{E}_{\lambda_{NL},\varepsilon}[\Sigma_1(i,i)]$;
  $\overline{\mathcal{G}}_4^k(:,\ :,\ i) = \mathbf{U}_1\tilde{\Sigma}_1\mathbf{V}_1$;
compute $\mathcal{G}_4^k = \text{ifft}\left(\overline{\mathcal{G}}_4^k,\ [\ ],\ 3\right)$.
Return $\mathcal{G}_4^k$

Equation (26) is just a function to ensure that the abundance is nonnegative where l_R+ is aiming at projecting the $\mathcal{A}$ on the nonnegative space. Therefore, it can be solved by

$$\mathcal{G}_5 \leftarrow \max(\mathcal{A}-\mathcal{H}_5, 0). \quad (42)$$

The role of (27) is to project the abundance vector to the affine space. According to [45], we can easily get the solution

$$\mathcal{G}_6 \leftarrow (\mathcal{A}-\mathcal{H}_6)+\text{repmat}(\mathbf{N}, I(1), 1))$$
$$\mathbf{N} = \frac{1}{P}[\mathbf{1}-\text{sum}((\mathcal{A}-\mathcal{H}_6))] \quad (43)$$

where $\mathbf{1}$ is an identity matrix. repmat$(\cdot)$ and sum$(\cdot)$ are both MATLAB functions.

4) Update all Lagrange multipliers $\mathcal{H}_1$, $\mathcal{H}_2$, $\mathcal{H}_3$, $\mathcal{H}_4$, $\mathcal{H}_5$, and $\mathcal{H}_6$

$$\mathcal{H}_1 = \mathcal{H}_1-\mathcal{A}_{\times 3}\mathbf{M}+\mathcal{G}_1$$
$$\mathcal{H}_2 = \mathcal{H}_2-\mathcal{A}+\mathcal{G}_2$$
$$\mathcal{H}_3 = \mathcal{H}_3-\mathcal{D}\mathcal{G}_2+\mathcal{G}_3$$
$$\mathcal{H}_4 = \mathcal{H}_4-\mathcal{A}+\mathcal{G}_4$$
$$\mathcal{H}_5 = \mathcal{H}_5-\mathcal{A}+\mathcal{G}_5$$
$$\mathcal{H}_6 = \mathcal{H}_6-\mathcal{A}+\mathcal{G}_6. \quad (44)$$

The complete iteration pseudocode of the NLTR algorithm proposed in this article is presented in Algorithm 2.

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Algorithm 2 Overall Pseudocode of NLTR Model

Input: The raw data $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, the number of endmembers P, the size of window q, the number of nonlocal patches K, and the parameters $\lambda_{TV} > 0$, $\lambda_{NL} > 0$

Output: $\mathcal{A}$, $\mathbf{M}$

Initialize $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times P}$ by VCA-based method, $\mathbf{M} \in \mathbb{R}^{P \times I_3}$ by FLCL, $\mathcal{G}_1$, $\mathcal{G}_2$, $\mathcal{G}_3$, $\mathcal{G}_4$, $\mathcal{G}_5$, $\mathcal{G}_6$, $\mathcal{H}_1$, $\mathcal{H}_2$, $\mathcal{H}_3$, $\mathcal{H}_4$, $\mathcal{H}_5$, $\mathcal{H}_6$

while *not convergence* **do**

  Update endmember $\mathbf{M}$;

  Update abundance tensor $\mathcal{A}$;

  Update six intermediate variables $\mathcal{G}_1$, $\mathcal{G}_2$, $\mathcal{G}_3$, $\mathcal{G}_4$, $\mathcal{G}_5$, and $\mathcal{G}_6$;

  Update Lagrange multipliers $\mathcal{H}_1$, $\mathcal{H}_2$, $\mathcal{H}_3$, $\mathcal{H}_4$, $\mathcal{H}_5$, and $\mathcal{H}_6$.

Return Endmeber matrix $\mathbf{M}$, Abundance tensor $\mathcal{A}$

IV. Experiments

In this section, we verify the effectiveness of the proposed algorithm NLTR on synthetic data and real data. We compare the existing five superior unmixing algorithms, namely, MV-NTF [6], ULTRA-V [33], NL-TSUn [36], SGSNMF [46], and SULRSR-TV [47]. Among them, MV-NTF and ULTRA-V are both proven effective algorithms based on tensor factorization. NMF-QMV, SGSNMF, and SULRSR-TV are all unmixing models based on the NMF framework, which uses a spatial–spectral combination information. In addition, SULRSR-TV also uses TV regularization to keep the spatial local smoothness of HSI.

Two common metrics are used to evaluate the performance of six algorithms, i.e., RMSE and SAD, in which RMSE is used to describe the signal-to-reconstruction error between the estimated abundance $\mathcal{A}$ and the true abundance $\hat{\mathcal{A}}$, and SAD is used to measure the spectral angular distance between the extracted endmember spectrum M and ground truth $\hat{\mathrm{M}}$. Formulaically, they can be described as

$$\begin{aligned} \mathrm{rmsE} &= \sqrt{\left|\mathcal{A} - \hat{\mathcal{A}}\right|^2} \\ \mathrm{SAD} &= \cos^{-1}\left(\frac{\hat{\mathbf{M}}^T \mathbf{M}}{\left\|\hat{\mathbf{M}}\right\| \|\mathbf{M}\|}\right). \end{aligned} \tag{45}$$

To ensure the fairness of the experimental comparison, we use VCA [3] to initialize the endmember matrix $\mathbf{M}$, and FCLS is utilized to estimate the abundance tensor $\mathcal{A}$. All algorithms are executed in MATLAB R2018b on a desktop computer with an Intel Core i5 (at 3.2 GHz) CPU and 16 GB of memory.

A. Experiments With Synthesized Datasets

In this section, we artificially synthesize two basic data. The endmember spectra are randomly selected from the USGS spectral library [48] and assumed to be pure. The specific synthesis process will be described in detail in the following.

1) *Dataset 1 (DS1):* Given the number of endmembers $P = 5$, the image size is 75×75, the number of bands is 224, and the spectrum is randomly selected from USGS spectral library. Then, DS1 is generated according to the method described in [47]. Specifically, each pixel follows a linear mixed model and satisfies the ANC and ASC constraints on the abundance map at the same time. Fig. 2(a) shows the five endmember spectra used to synthesize the data. The abundance map corresponding to each endmember is presented in Fig. 2(b)–(f).

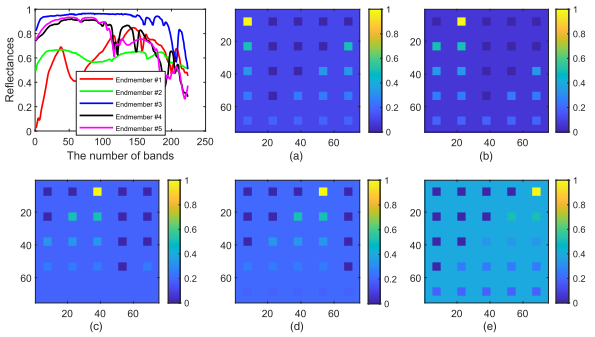


Fig. 2. Simulated DS1. (a) Endmembers spectra. (b) Endmember 1. (c) Endmember 2. (d) Endmember 3. (e) Endmember 4. (f) Endmember 5.

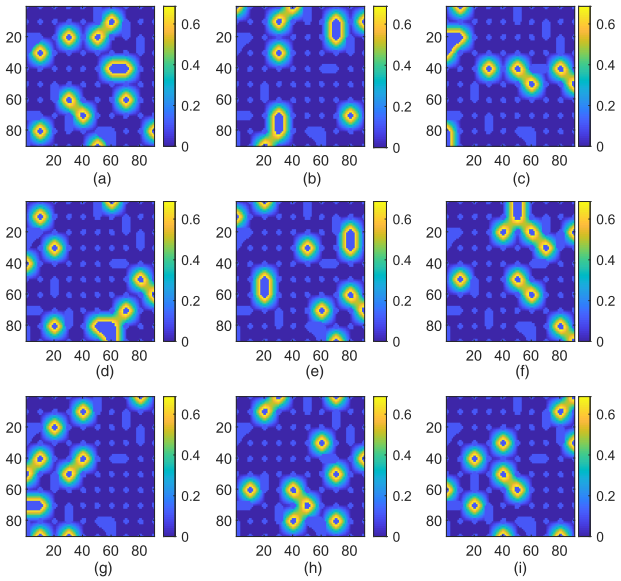


Fig. 3. Simulated DS2. (a) Endmember 1. (b) Endmember 2. (c) Endmember 3. (d) Endmember 4. (e) Endmember 5. (f) Endmember 6. (g) Endmember 7. (h) Endmember 8. (i) Endmember 9.

2) *Dataset 2 (DS2):* DS2 contains nine spectra randomly selected from the USGS library, with a size of 90×90. The ANC and ASC constraints of abundance are

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TABLE I
RMSE RESULTS ON THE SYNTHETIC DATA

Data	SNR	SGSNMF	ULTRA-V	SULRSR-TV	NL-TSUn	MV-NTF	NLTR
DS1	10 dB	0.0946	0.2339	0.0995	0.1447	0.1906	**0.0832**
	20 dB	0.0944	0.1701	0.0805	0.1024	0.1500	**0.0425**
	30 dB	0.1227	0.2291	0.0797	0.0828	0.1456	**0.0127**
DS2	10 dB	0.1276	0.1980	0.1433	0.1326	0.1256	**0.1098**
	20 dB	0.1189	0.1876	0.1366	0.1273	0.1580	**0.0404**
	30 dB	0.1085	0.1690	0.1344	0.0987	0.1005	**0.0101**

TABLE II
SAD RESULTS ON THE SYNTHETIC DATA

Data	SNR	SGSNMF	ULTRA-V	SULRSR-TV	NL-TSUn	MV-NTF	NLTR
DS1	10 dB	0.3081	0.1427	0.1115	0.1326	0.5127	**0.1024**
	20 dB	0.0626	0.1214	0.0453	0.0952	0.4023	**0.0225**
	30 dB	0.0427	0.0737	0.0134	0.0623	0.3210	**0.0078**
DS2	10 dB	0.2919	0.3172	0.1977	0.4329	0.5232	**0.1312**
	20 dB	0.2458	0.3202	0.1688	0.4061	0.5161	**0.0538**
	30 dB	0.2451	0.3110	0.1423	0.2846	0.3237	**0.0127**

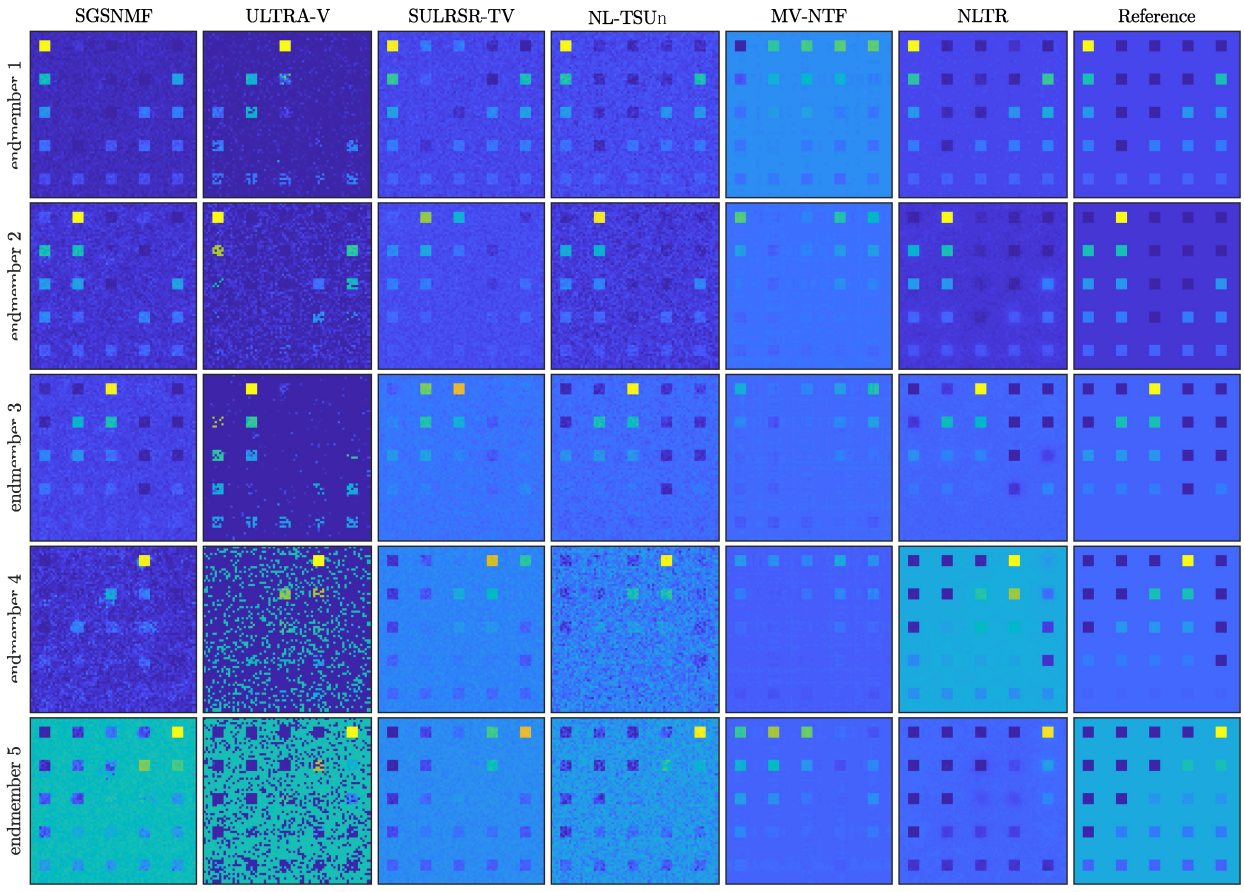


Fig. 4. Abundance maps estimated by the five comparison algorithms and the proposed NLTR method in DS1 with 20 dB.

also enforced. Different from DS1, the construction method of DS2 is similar to [49]. Fig. 3 shows nine kinds of abundance maps, respectively. It can be seen from the figure that the abundance has smoothness in the spatial domain.

After two data are successfully constructed, the Gaussian noise is added to them with different SNRs, respectively, i.e., 10, 20, and 30 dB. Then, the validity of the proposed algorithm and the other five comparison algorithms is verified on the six synthetic data. For the fairness of the experiments,

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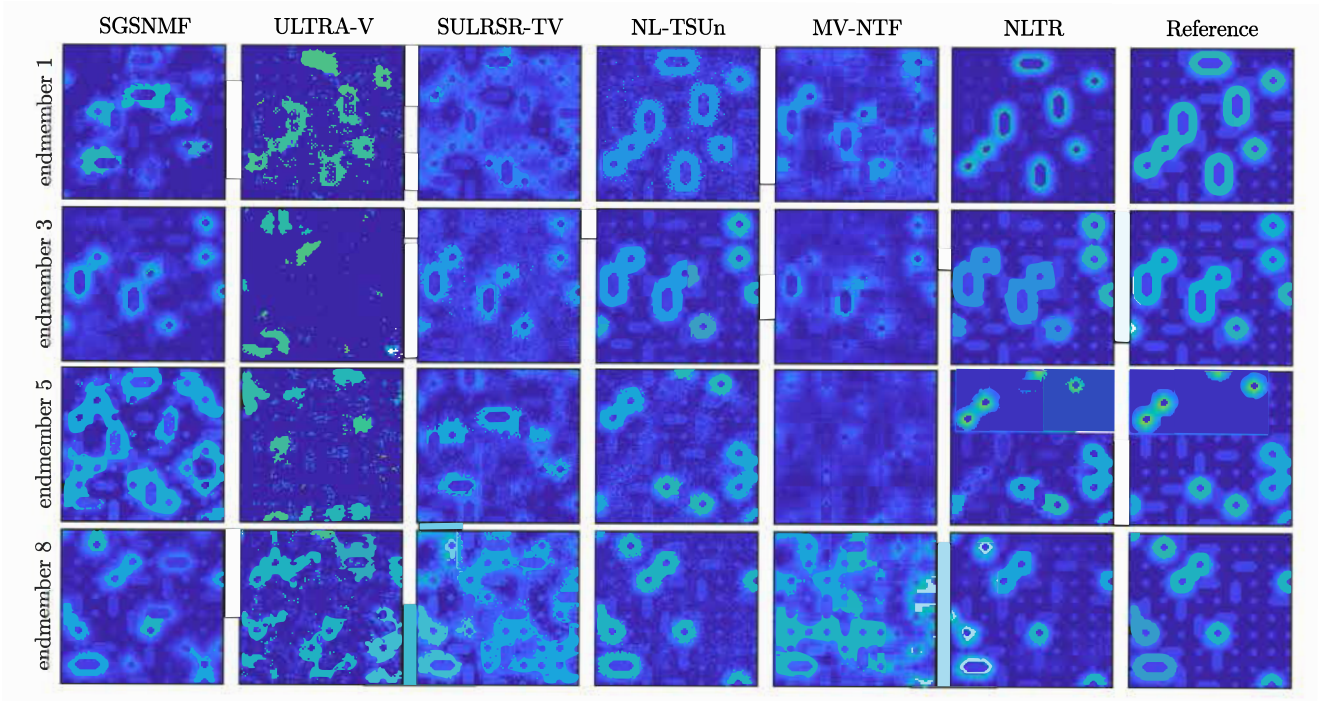


Fig. 5. Abundance maps of endmember 1, endmember 3, endmember 5, and endmember 8, estimated by the five comparison algorithms and the proposed NLTR method in DS2 with 20 dB.

we use Monte Carlo sampling for each algorithm and take the average after running 20 times as the final result.

Tables I and II show the RMSE and SAD results of the six data on the five comparison algorithms and the proposed method NLTR, respectively. From them, we can observe that the performance of MV-NTF on each database is poor. It may be because MV-NTF only uses the spectral information of data but ignores spatial relationships. SGSNMF and NMF-QMV, which are both based on NMF, have a relatively good unmixing performance on DS1 and DS2. However, they have poor performance at low SNR. SULRSR-TV performs better on DS1, which is consistent with the experimental results described in their original article. Unfortunately, they have poor results on the DS2 data that we designed, most likely because there are no pure pixels, and endmembers are highly mixed. At the same time, since ULTRA-V focuses on the endmember variability, and there is no variability in synthetic data, this method performs the worst.

The above analyses have also been verified in the visualization maps of abundance. Specifically, as shown in Fig. 4, the DS1 data are unmixed by six algorithms, and the abundance maps are all displayed. It can also be seen that the proposed NLTR has the best performance in estimating abundance and is less affected by noise. DS2 has a large number of endmembers; hence, we select the abundance maps corresponding to four endmembers to show. It can be concluded from Fig. 5 that our method is the least affected by noise, and ULTRA-V also performs poorly here.

B. Experiments With Real Datasets

In this section, we choose three datasets to evaluate the performance of the proposed NLTR: Jasper, Samon, and Cuprite.

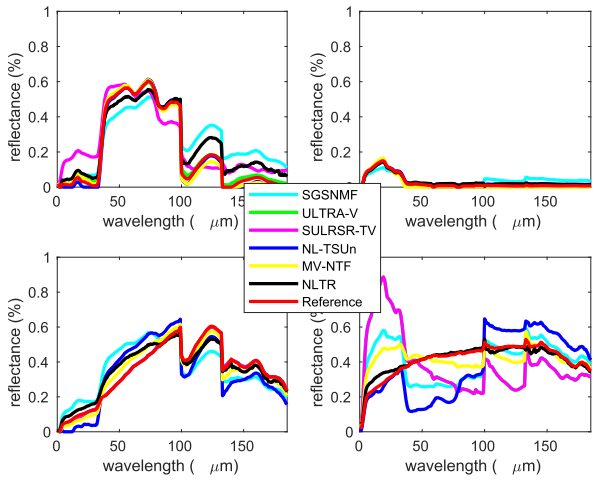


Fig. 6. Endmembers spectra of the Jasper Ridge dataset for all tested method, and the black line is ground truth.

These three datasets are all captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), which has different image sizes and numbers of endmembers. The number of bands in the three datasets is 224. After removing the bands polluted by water vapor and noise, the numbers of bands in each database are different. The difference of these parameters can fully reflect the adaptability of the NLTR algorithm. The specific content will be described in detail in the following.

1) Jasper Ridge Dataset: The wavelength range of Jasper data is 380–2500 nm, with 10-nm spectral resolution. The original image size is 512 × 614, and we only select one of the 100 × 100 sizes. After removing the useless bands, the final number of bands is set to 185. The image contains four

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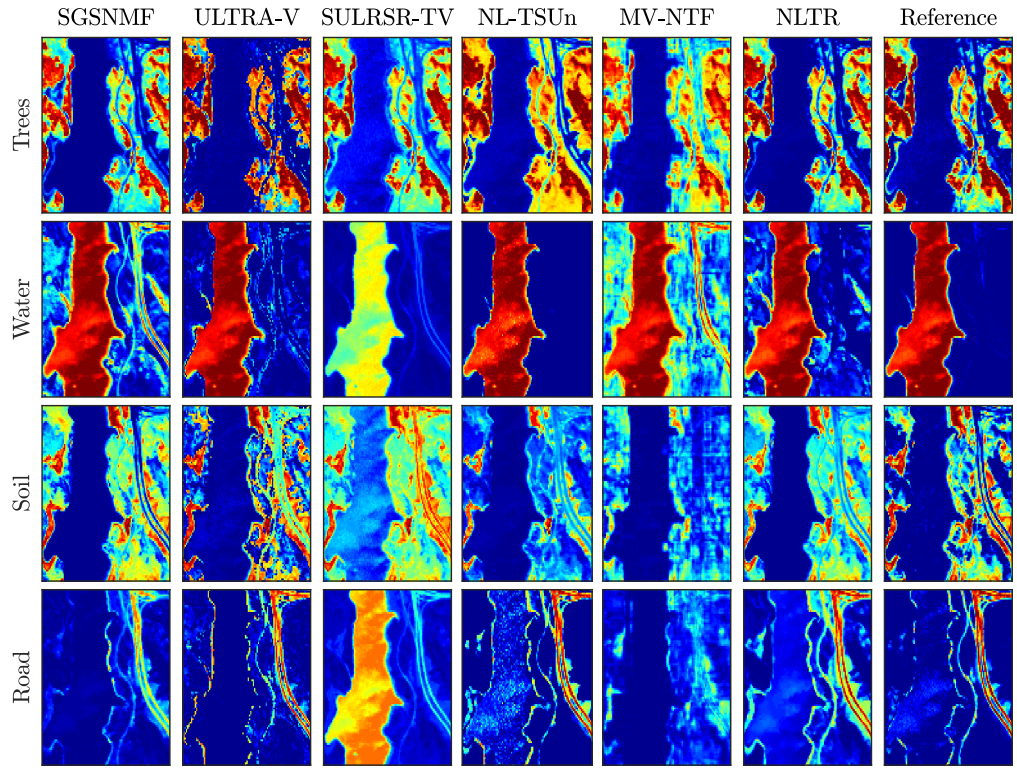


Fig. 7. Abundance maps of the Jasper Ridge dataset for all tested method.

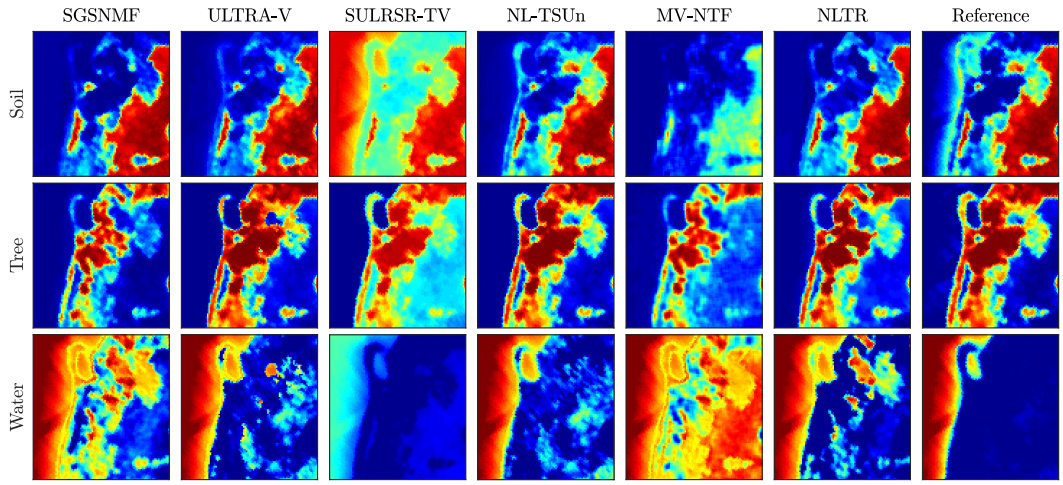


Fig. 8. Abundance maps of the Samon dataset for all tested method.

TABLE III
SAD RESULTS ON THE JASPER RIDGE DATA

	SGSNMF	ULTRA-V	SULRSR-TV	NL-TSUn	MV-NTF	NLTR
Tree	0.2069	**0.1774**	0.1783	0.2081	0.2512	0.2613
Water	0.2063	0.4246	0.2329	0.4929	0.2479	**0.0793**
Soil	0.1317	0.1499	**0.1302**	0.2334	0.1703	0.1614
Road	0.0555	0.2998	0.5109	0.3665	0.3889	**0.0437**
Mean	0.1626	0.2846	0.3017	0.3723	0.2760	**0.1364**
RMSE	0.1559	0.1212	0.2672	0.1192	0.2446	**0.1128**

TABLE IV
SAD RESULTS ON THE SAMSON DATA

	SGSNMF	ULTRA-V	SULRSR-TV	NL-TSUn	MV-NTF	NLTR
Tree	**0.0098**	0.0344	0.0207	0.0288	0.0583	0.0465
Water	0.0519	0.0584	**0.0495**	0.0496	0.0921	0.0647
Soil	0.2151	0.2470	0.1299	0.1289	0.2573	**0.0937**
Mean	0.1279	0.1479	0.0812	0.0825	0.1613	**0.0710**
RMSE	0.2782	0.1442	0.2962	**0.0934**	0.3913	0.1024

endmembers: soil, water, tree, and road. The ground truth is obtained as the same as [6]. The SAD results of our proposed algorithm NLTR and the other five comparison algorithms on Jasper data are shown in Table III. The comparison between the spectral curve and the ground truth of each endmember is shown in Fig. 6. In addition, Fig. 7 shows the abundance maps of the four endmembers. Although our results are not optimal

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TABLE V
SAD RESULTS ON THE AVIRIS CUPRITE DATA

	SGSNMF	ULTRA-V	SULRSR-TV	NL-TSUn	MV-NTF	NLTR
Alunite	0.1842	0.1897	0.0966	0.0783	0.0748	**0.0735**
Andradite	0.0682	0.0971	0.1353	0.0958	**0.0661**	0.0940
Buddingtonite	0.0954	0.1097	0.0699	0.0982	0.0718	**0.0698**
Chalcedony	0.1525	0.1212	0.1482	0.1253	0.1652	**0.0761**
Dumortierite	0.1014	0.0936	**0.0855**	0.1709	0.0984	0.1162
Kaolinite#1	0.0790	0.1300	0.0718	0.0738	0.0841	**0.0679**
Kaolinite#2	0.1020	**0.0391**	0.0678	0.0625	0.0786	0.0491
Montmorillonite	0.0543	0.0508	**0.0504**	0.0720	0.0535	0.0582
Muscovite	**0.0788**	0.1257	0.1066	0.0925	0.2518	0.1449
Nontronite	**0.0704**	0.0835	0.0730	0.0813	0.1086	0.0730
Pyrope	0.0732	0.1737	0.1876	0.1501	0.1431	**0.0588**
Sphene	0.2261	0.2306	0.0663	0.1642	0.0700	**0.0554**
Mean	0.1185	0.1317	0.1042	0.1054	0.1176	**0.0825**

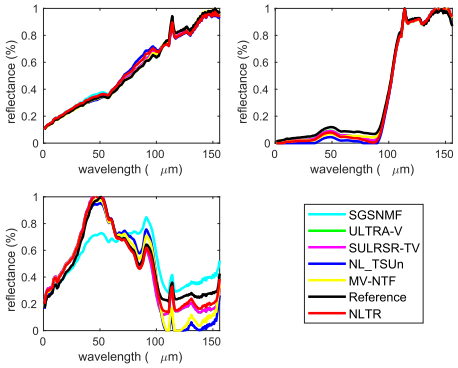


Fig. 9. Endmembers spectra of the Samon dataset for all tested method, and the black line is ground truth.

for each endmember, it can be seen that our proposed method NLTR is significantly better than the other five comparison algorithms in terms of visual effects and specific quantitative indicators.

2) Samon Dataset: The image size of Samon data used in this experiment is 95 × 95. There are 156 bands retained in the data, and it has three endmembers: soil, tree, and water. Fig. 8 shows the abundance map estimated by six algorithms, which also shows that NLTR is closer to ground truth. As shown in Fig. 9, the endmember spectra extracted by the six algorithms are drawn correspondingly. It can be seen that the algorithms generally have a large error on the soil endmember, which may be caused by a large number of other substances mixed in the soil, which is caused by the nonlinear photon mixing during the data transmission. Even so, the proposed algorithm NLTR also performs better. Table IV shows the specific quantification results of each method on SAD. It can be seen that, although NLTR does not perform.

3) Cuprite Dataset: Cuprite is widely used in unmixing experiments. These data are mining areas, and the various minerals contained are in a highly mixed state. A part of the raw data is cropped out for this experiment, in which image size is 200 × 150, and 188 bands are retained, and there are 12 pure endmembers. For Cuprite data, the ground truth of endmembers and abundances is not provided by the laboratory. Therefore, we determine the reference endmember according to [50]. The SAD results of the six methods are presented in Table V. It can be concluded that the average SAD of NLTR is smaller than the other five comparison algorithms, and the six endmember spectra are all optimal. The spectral curves extracted by six algorithms are also drawn in Fig. 10. Combining the above results, it can be concluded that NLTR has a better performance no matter it is a simple database or a complex database.

C. Discussion

In this section, we mainly discuss the parameters involved in the proposed method NLTR. Specifically, NLTR mainly involves four parameters: the nonlocal low-rank tensor regularization parameter λ_{NL}, the local smooth regularization λ_{TV}, the number of clusters K, and the size of nonlocal patch q. Among these parameters, when λ_{NL} controls the nonlocal spatial similarity in the image, λ_{TV} is used to control the local smoothness in abundance maps. To judge these parameters, we take DS1 with an SNR of 20 dB as an example to illustrate how to set the parameters in our experiments.

First, the influence of the number of clusters K on the experimental results is analyzed. We set the size of λ_{NL}, λ_{TV}, and patch size to 0.01, 0.01, and 3, respectively. The numbers of clusters K are defined as {10, 20, 30, 40, 50, 60, 70, 80, 90, 100}. The obtained SAD and RMSE values are shown in Fig. 11. It can be seen that the two evaluation indicators RMSE and SAD have a small floating range with a change of K, but, when K is set 50, the best result is still obtained. This is because, if K is too small, many patches with big differences will be grouped together, and if K is too large, the effect of nonlocal similarity constraints will be weakened.

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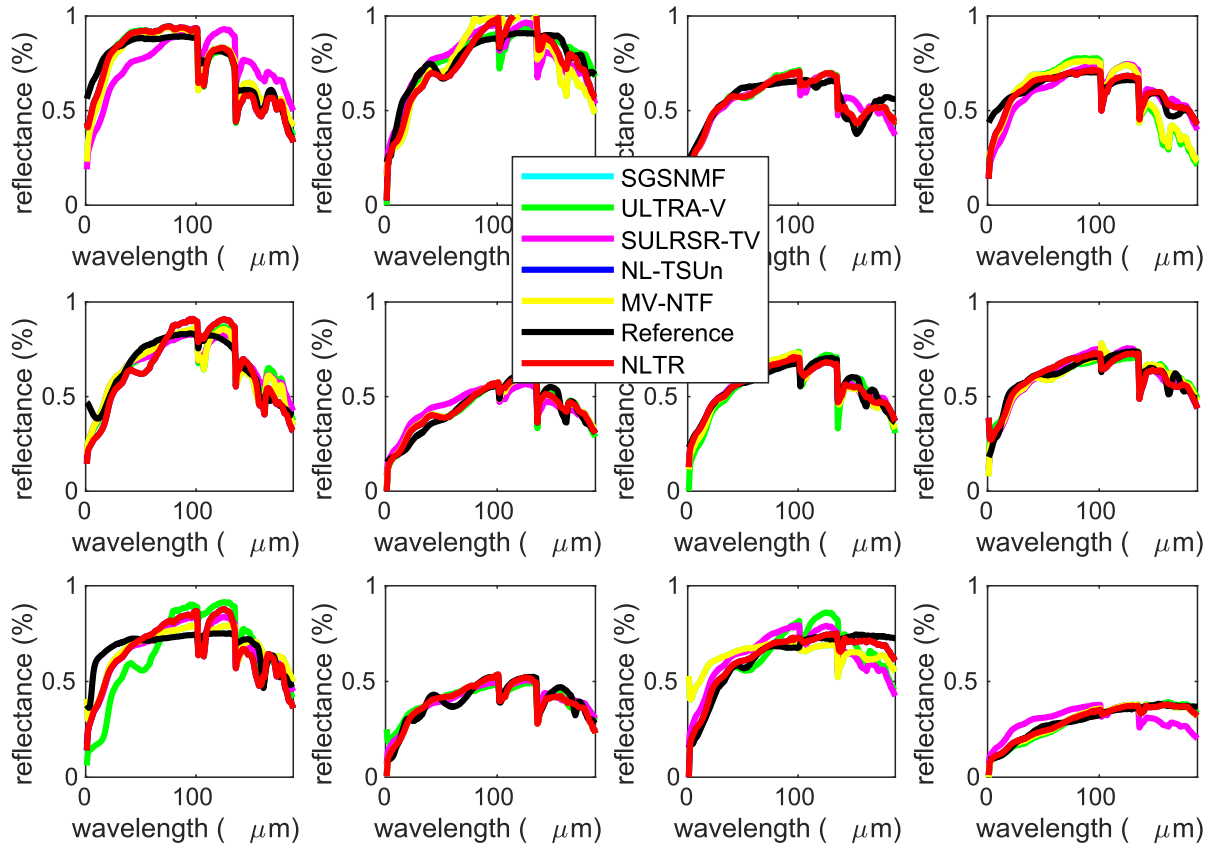


Fig. 10. Endmembers spectra of the Cuprite dataset for all tested method, and the black line is ground truth.

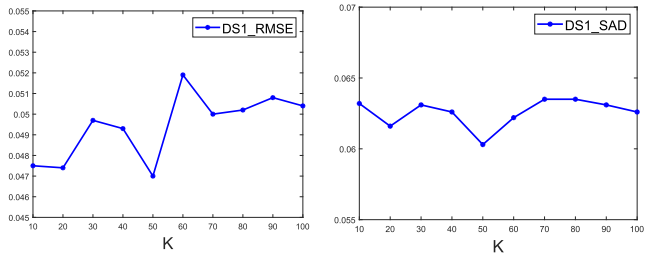


Fig. 11. Comparison of NLTR with different clustering numbers in terms of (a) RMSE and (b) SAD.

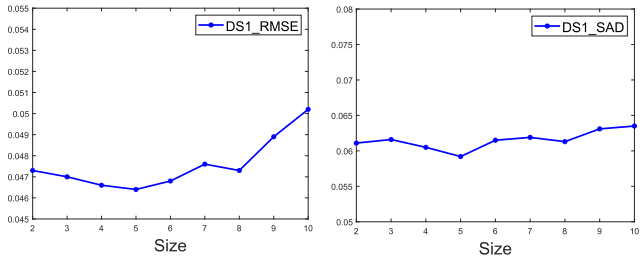


Fig. 12. Comparison of NLTR with different patch sizes in terms of (a) RMSE and (b) SAD.

Subsequently, we determine K as 50, and keep λ_{NL}, λ_{TV} value are both 0.01. The sizes of patches are set to {2, 3, 4, 5, 6, 7, 8, 9, 10}, respectively. The obtained SAD and RMSE values are drawn in Fig. 12. It can be concluded that, when $q = 5$, RMSE and SAD are taken to the minimum, which shows that the unmixing effect is best at this point. As the patch size increases, the unmixing performance continues to decline. To a large extent, it is because the larger the patch, the more materials it contains, and the harder it is to find the most similar patch in the nonlocal low-rank constraint. On the other hand, when the patch is too small, it is not easy to fully explore the nonlocal similarity in the image.

Finally, we discuss the influence of two regularization parameters λ_{NL} and λ_{TV} on the proposed NLTR. At this time, K is fixed as 50, and the patch size is 5. Let λ_{NL} and λ_{TV} are set to {0, 5e−3, 1e−2, 1.5e−2, 2e−2, 2.5e−2, 3e−2, 3.5e−2, 4e−2, 4.5e−2, 5e−2}. The unmixing results obtained are shown in Fig. 13. As can be seen from Fig. 13,

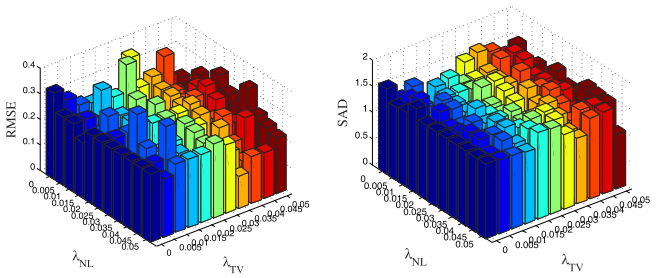
TABLE VI
PARAMETER SETTINGS

Data		K	q	λ_{NL}	λ_{TV}
DS1	10 dB	50	3	0.05	0.01
	20 dB	50	5	0.01	0.02
	30 dB	50	5	0.02	0.005
DS2	10 dB	30	3	0.01	0.005
	20 dB	30	3	0.001	0.01
	30 dB	30	3	0.01	0.05
Jasper		50	5	0.001	0.01
Samon		50	5	0.001	0.01
Cuprite		50	5	0.001	0.01

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TABLE VII
RUNNING TIME (SECONDS) OF ALL ALGORITHMS ON THREE REAL DATASETS

	SGSNMF	ULTRA-V	SULRSR-TV	NL-TSUn	MV-NTF	NLTR
Cuprite Data	187	209	34	194	904	116
Jasper Ridge Data	41	26	6	55	170	45
Samon Data	37	14	5	29	135	35



(a) (b)

Fig. 13. (a) RMSE and (b) SAD versus the parameters λ_{NL} and λ_{TV} of the proposed method on DS1 with 20 dB.

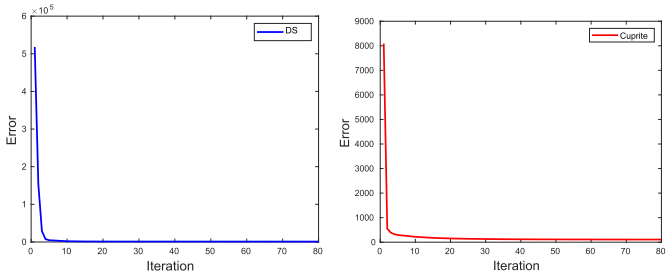


Fig. 14. Convergence analysis of NLTR on DS1_20 and Cuprite data.

under the simultaneous action of two parameters λ_{NL} and λ_{TV}, the unmixing performance of NLTR changes greatly. When λ_{NL} or λ_{TV} is 0, it means that the λ_{NL} or λ_{TV} constraint has no effect. It can also be seen from Fig. 13 that both λ_{NL} and λ_{TV} regularizations can improve the unmixing performance, which also verifies the effectiveness of the proposed method. In this experiment, considering the RMSE and SAD results, λ_{NL} and λ_{TV} are determined to 0.01 and 0.02, respectively. In summary, the parameter setting results of all experiments in this article are presented in Table VI.

Here, the time complexity of all algorithms is compared on three real datasets. Specifically, all experiments are performed on a Linux system with Intel Xeon CPU E5-2697 v3 at 2.60 GHz and 64-GB RAM. Table VII shows the results of the comparison. It can be summarized from Table VII that the time complexity of MV-NTF will explode as the image size increases. Our algorithm can converge well in a short period of time. In addition, the algorithm convergence analysis is carried out on Cuprite data and synthetic data DS1 with 20 dB, and the results are shown in Fig. 14. It can be seen that the proposed algorithm can converge very well both on synthetic data and real data, and the convergence speed is very fast.

V. CONCLUSION

In this article, we propose an algorithm called NLTR for hyperspectral unmixing. HSI is regarded as a 3-D tensor that directly processes the cube, which can preserve all information of data as much as possible. NLTR designs a nonlocal low-rank tensor regularization, which can be used to explore the nonlocal similarity of data. In addition, the TV regularizer is utilized to preserve the local smooth of abundance maps. The combination of nonlocal low-rank and TV regularization and applied to the NTF framework simultaneously can not only fully preserve the original information without any loss but also extract spatial information of data. Finally, NLTR is optimized by ADMM, and effective verification is carried out on multiple synthetic data and real data.

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